NOC N2/N3 | Network Incident Response Playbook

Technical paper • Wireshark / tshark + Zabbix + Grafana + Ansible | Version 1.0 | 01 November 2026

Executive summary. This paper defines a repeatable, evidence-driven workflow for ten network alerts. Zabbix detects and routes events, Grafana supports trend analysis, Wireshark/tshark provides packet-level evidence, and Ansible performs read-only reachability checks. It separates observation from remediation and requires change approval for production modifications.

1. Architecture and responsibilities

Zabbix: SNMP interface counters, ICMP reachability, trigger dependencies and incident tags. **Grafana:** timelines of latency, packet loss, utilization, discards and error rates, aligned by timestamp and site. **Wireshark:** authorized SPAN/TAP or existing PCAP inspection for retransmissions, duplicate ACK, zero-window, resets and protocol anomalies. **Ansible:** controller-side ICMP sampling and JSON/CSV evidence. Custom jitter and TCP retransmission percentages require additional collectors; do not assume native SNMP provides them.

2. Incident response lifecycle

• Detect: register alert ID, host, interface, site, severity, first occurrence, SLA and customer impact.

• Triage (N1/N2): confirm alert is not maintenance, administrative shutdown, polling failure or a false positive. Check correlated devices and upstream dependencies.

• Scope: compare site-wide versus isolated impact, redundancy state, BGP/OSPF adjacency, service path and recent changes.

• Collect: export Zabbix event timeline, Grafana panels, device interface counters, syslogs and authorized packet captures; record UTC timestamps.

• Diagnose: test at multiple points, distinguish physical faults from congestion, routing, application slowness or filtering.

• Contain/escalate: N2 contacts N3/vendor according to impact; no unsanctioned configuration changes.

• Recover: execute approved change, observe stabilization window, compare baseline, capture before/after evidence.

• Close: document root cause (confirmed or provisional), incident timeline, ticket, owner and preventive action.

3. Ten alarm-specific runbooks

01 Interface Down

Signal: Operational status down for 2 min while administrative status is up.
Diagnosis: Verify ifAdminStatus, physical light, optics, interface logs and peer state; compare redundant paths.
Escalation: N2 checks cable/SFP/peer; N3/vendor for widespread or recurrent outage.
Recovery criteria: Interface up, error-free and service restored for agreed stabilization period.

02 CRC / FCS Errors

Signal: More than 100 input errors in 5 min, tuned to link baseline.
Diagnosis: Read ifInErrors and vendor CRC/FCS counters; check optics Rx/Tx, duplex, cabling and port history.
Escalation: N3/field support for cable or transceiver replacement under change control.
Recovery criteria: Error delta returns to baseline after repair.

03 Packet Loss

Signal: ICMP loss greater than 5% over 5 min.
Diagnosis: Probe source, next hop and destination; compare paths; verify ACL, CoPP and ICMP rate limiting.

Escalation: N3 for confirmed service packet loss or multi-hop impact.
Recovery criteria: End-to-end loss below agreed SLA with application confirmation.

04 High Latency

Signal: Average RTT greater than 100 ms for 5 min.
Diagnosis: Check RTT by hop and site, queueing, utilization, path change, CPU and asymmetric routes.
Escalation: N3/carrier when path or congestion cause confirmed.
Recovery criteria: RTT sustained under baseline/SLA; no service degradation.

05 Bandwidth Saturation

Signal: Interface utilization greater than 85% for 10 min.
Diagnosis: Compare in/out bit rates with ifHighSpeed, 95th percentile, burst patterns, QoS and drops.
Escalation: Capacity planning or N3 QoS change with approval.
Recovery criteria: Sustained utilization within headroom and queue drops controlled.

06 Interface Discards

Signal: More than 100 discards in 5 min.
Diagnosis: Separate in/out discards; inspect buffers, QoS queues, microbursts and policers.
Escalation: N3 for congestion or QoS tuning.
Recovery criteria: Discards stable at baseline with throughput intact.

07 Network Jitter

Signal: Jitter over 30 ms measured by active probe.
Diagnosis: Verify probe interval, clock, voice/video path, packet delay variation and competing traffic.
Escalation: N3/QoS owner for real-time traffic impact.
Recovery criteria: Jitter within voice/video SLA on repeat tests.

08 TCP Retransmissions

Signal: Retransmission ratio over 3% in defined monitored flow population.
Diagnosis: Inspect tcp.analysis.retransmission, duplicate_ack, zero_window, RTT and capture point; correlate server/client metrics.
Escalation: N3/application team for transport or endpoint bottleneck.
Recovery criteria: Retransmission ratio normal with application recovery.

09 Interface Flapping

Signal: Three or more operational state transitions in 10 min.
Diagnosis: Review link up/down logs, power, optics, LACP, STP, peer and hardware events.
Escalation: N3/field team for intermittent hardware or link fault.
Recovery criteria: No repeated transitions during stabilization window.

10 Critical Network Degradation

Signal: Packet loss >5%, RTT >100 ms and utilization >85% on aligned path/window.
Diagnosis: Correlate all three series, discard/CRC counters, routing changes and affected applications; rule out monitoring artifacts.
Escalation: Immediate N3/incident commander; carrier/vendor if shared link affected.
Recovery criteria: All three indicators and business service return to SLA; postmortem required.

4. Packet analysis reference

Wireshark display filters: `tcp.analysis.retransmission`; `tcp.analysis.duplicate_ack`; `tcp.analysis.zero_window`; `tcp.flags.reset == 1`; `icmp`. Filters indicate symptoms, not standalone proof of root cause. Capture position and packet loss at the capture point can affect interpretation.

5. Severity, ownership and evidence

Warning: N2 investigates within the operational response window. **High:** N2 initiates incident, alerts N3 on impact or persistence. **Disaster:** incident commander and N3 engaged immediately under local policy. Retain event ID, graphs, timestamps, topology, CLI snapshots, PCAP hashes, approved change ID, before/after tests and owner. Do not include credentials or unrestricted sensitive payloads in tickets.

6. Automation execution and boundaries

Edit `inventory/targets.yml`, then run `ansible-playbook -i inventory/targets.yml playbooks/diagnostics.yml`. Inspect `reports/summary.csv` and `reports/icmp_raw.json`. The executable automation currently tests packet loss and mean RTT only. Other eight alarms are documented operational procedures, not implemented collectors. Offline PCAP triage is optional: `bash scripts/pcap_triage.sh capture.pcapng reports/pcap`.

7. Acceptance tests and improvement plan

Validate with authorized test destinations: reachable host, blocked ICMP, induced latency in lab, 100% loss, missing ping binary, malformed PCAP, and repeated runs. Confirm no configuration changes, reports are private (0600/0700), results are timestamped, and UNKNOWN is not confused with OK. Next phase: Zabbix API read-only event ingestion, SNMPv3 counters, IP SLA jitter, dedicated TCP telemetry, Grafana datasource provisioning, and CI linting.

Conclusion. A robust NOC response requires time-aligned telemetry, clear escalation, reproducible evidence and explicit recovery criteria. The included Ansible tool is intentionally limited and read-only; it complements rather than replaces Zabbix monitoring and engineering judgement.

References: Zabbix documentation (triggers and SNMP); Wireshark User Guide (display filters); Grafana documentation (dashboards); Ansible documentation (builtin.command and playbooks); RFC 2863 (Interfaces Group MIB); RFC 3393 (IP Packet Delay Variation).